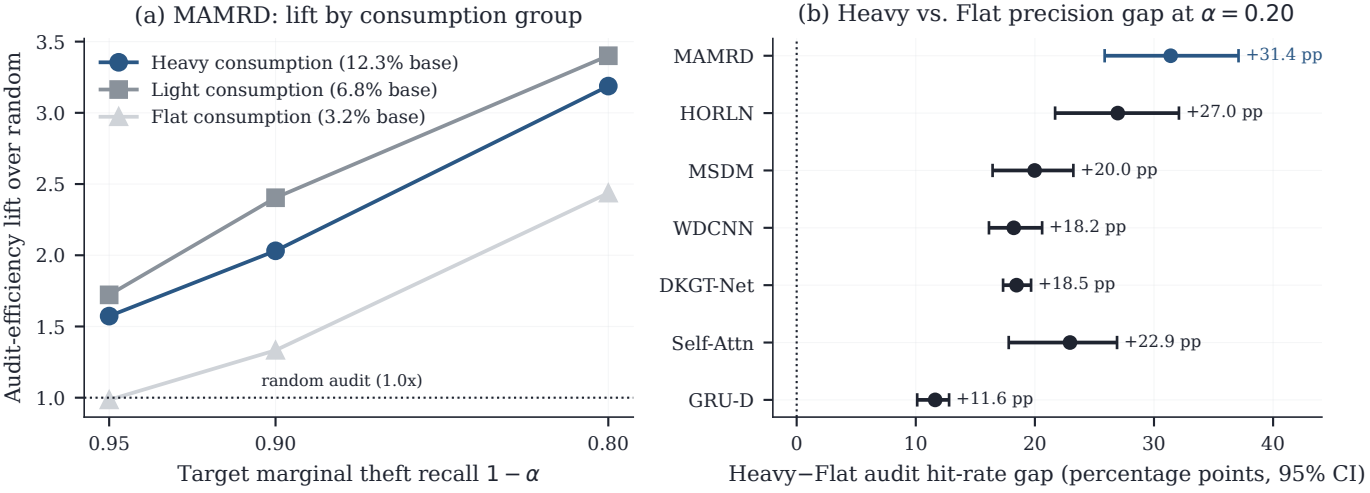
(a) MAMRD: lift by consumption group
Heavy consumption (12.3% base)
Light consumption (6.8% base)
Flat consumption (3.2% base)
Audit-efficiency lift over random
3.5
3.0
2.5
2.0
1.5
1.0
random audit (1.0x)
0.95
0.90
0.80
Target marginal theft recall 1 − α
(b) Heavy vs. Flat precision gap at α = 0.20
MAMRD
HORLN
MSDM
WDCNN
DKGT-Net
Self-Attn
GRU-D
+31.4 pp
+27.0 pp
+20.0 pp
+18.2 pp
+18.5 pp
+22.9 pp
+11.6 pp
0
10
20
30
40
Heavy−Flat audit hit-rate gap (percentage points, 95% CI)